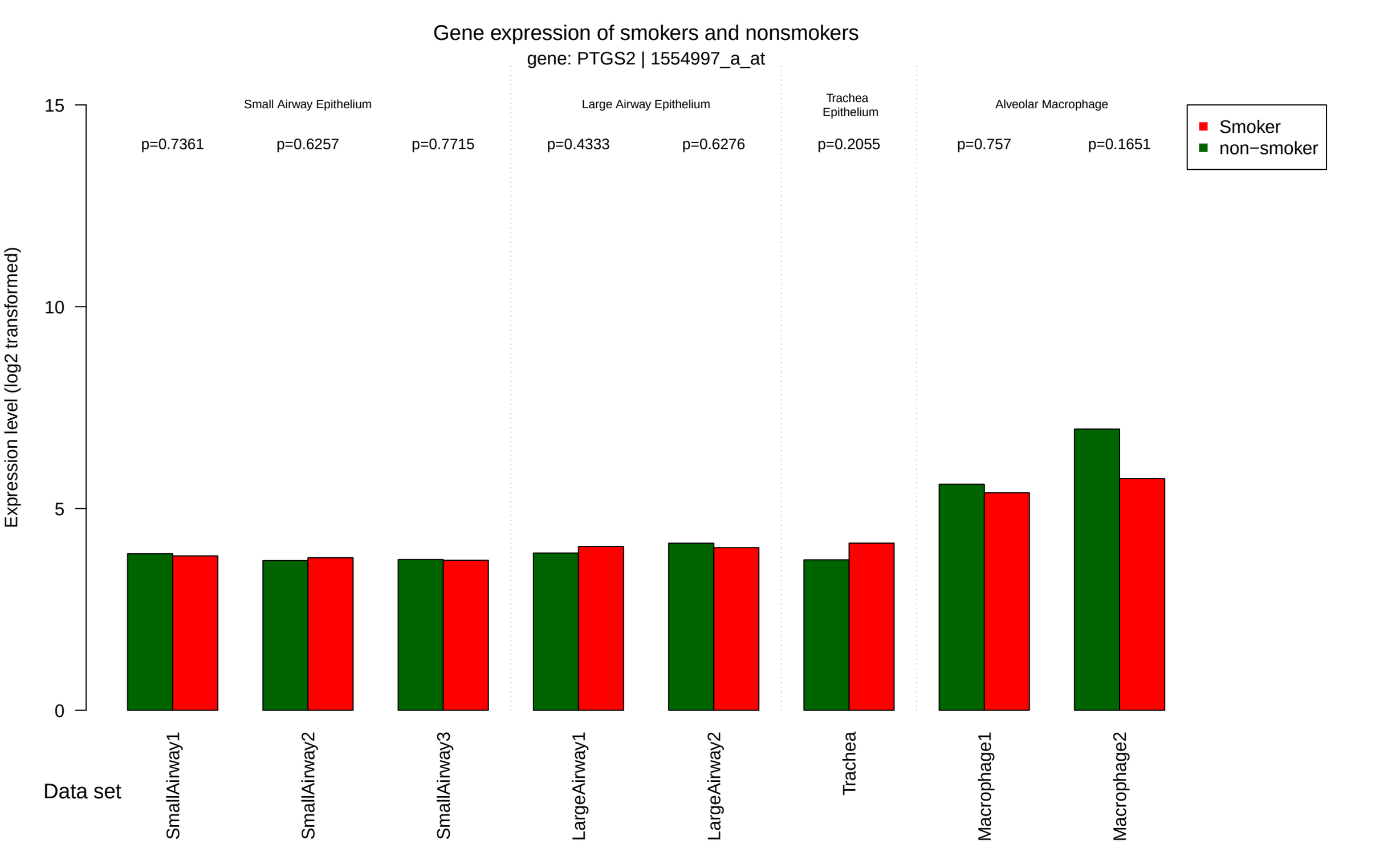
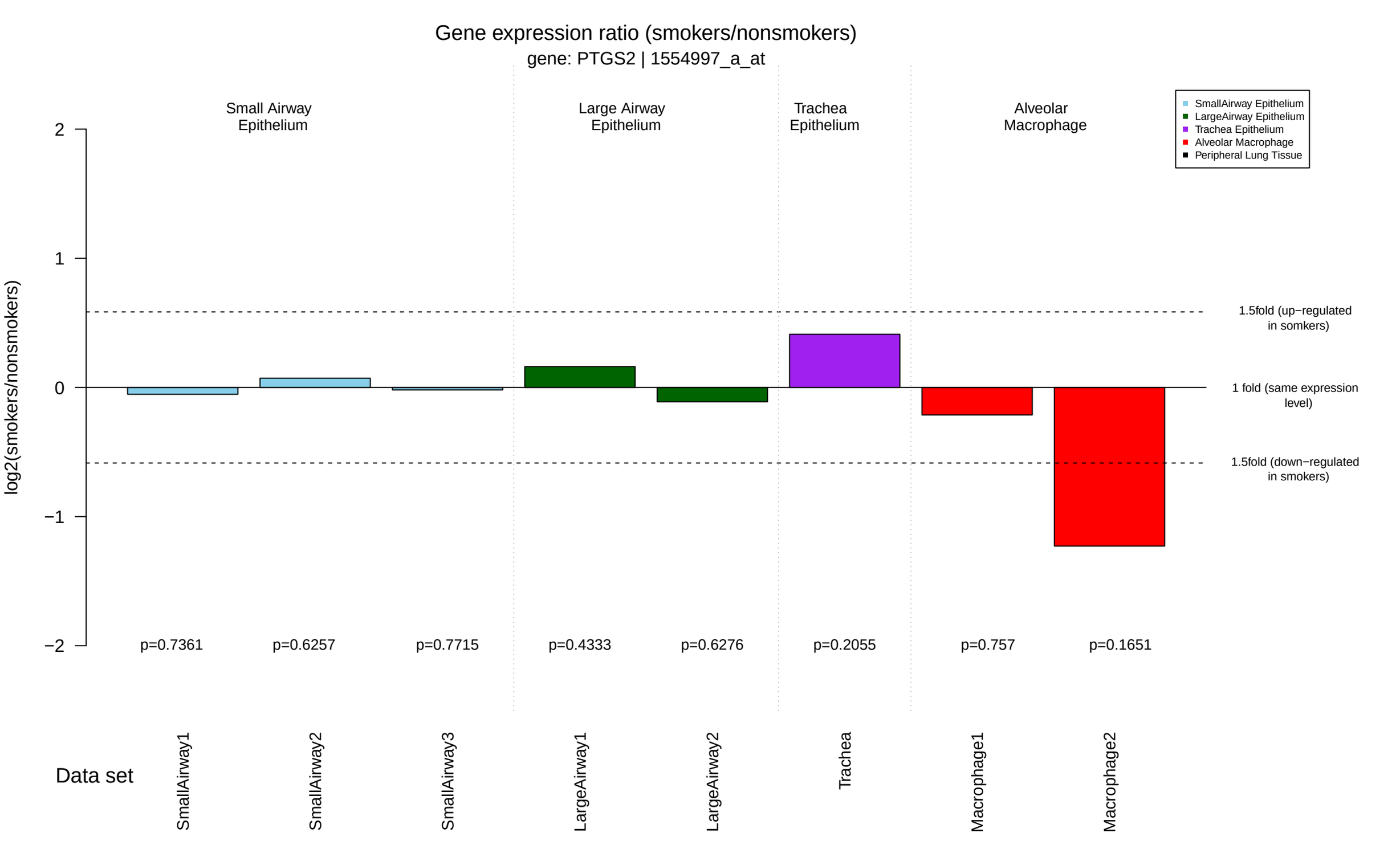
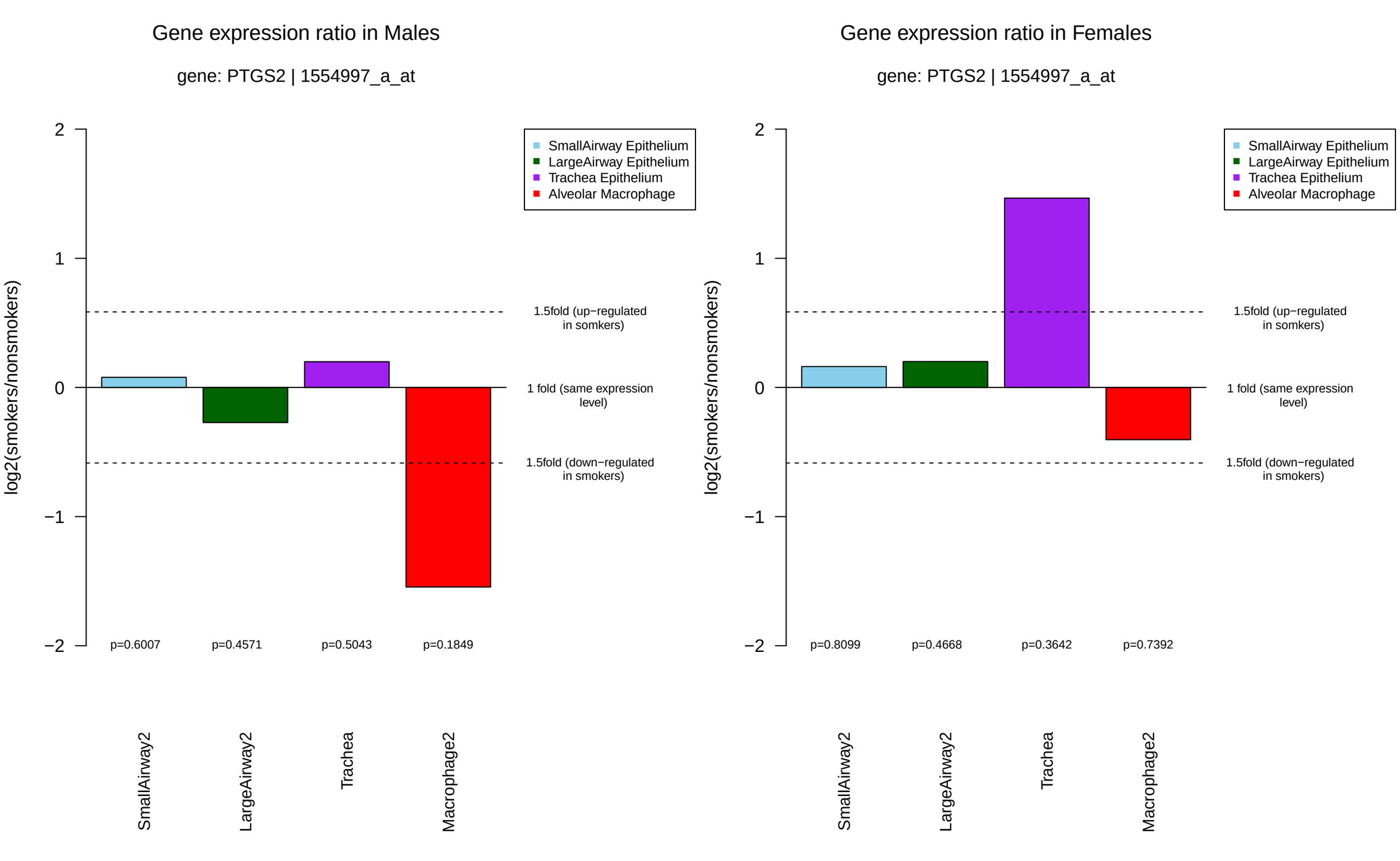
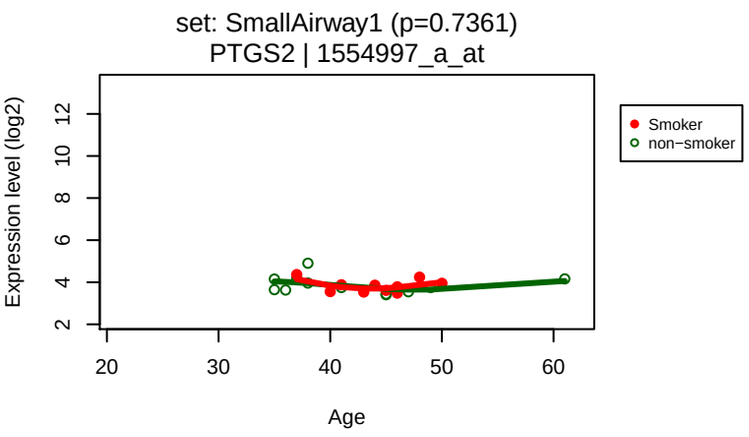
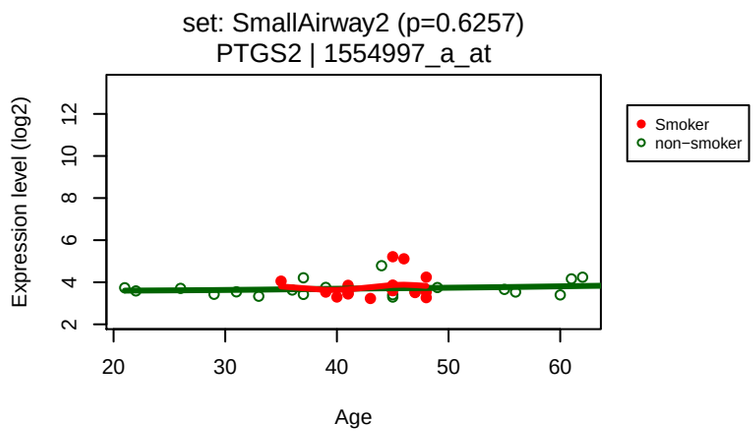
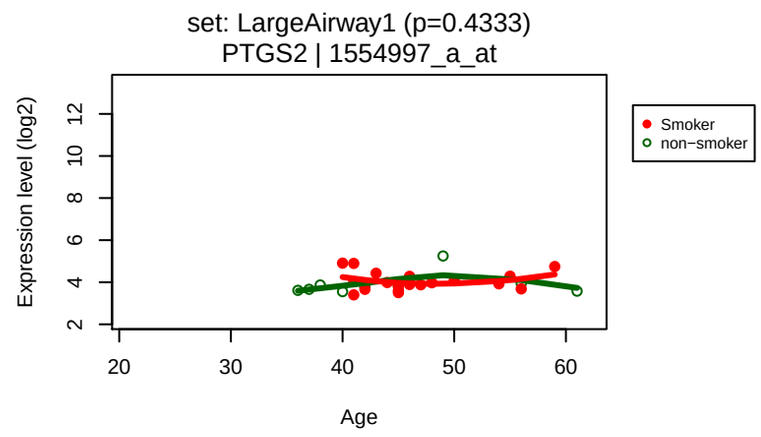

Gene expression of smokers and nonsmokers
gene: PTGS2 | 1554997_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7361
p=0.6257
p=0.7715
p=0.4333
p=0.6276
p=0.2055
p=0.757
p=0.1651
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

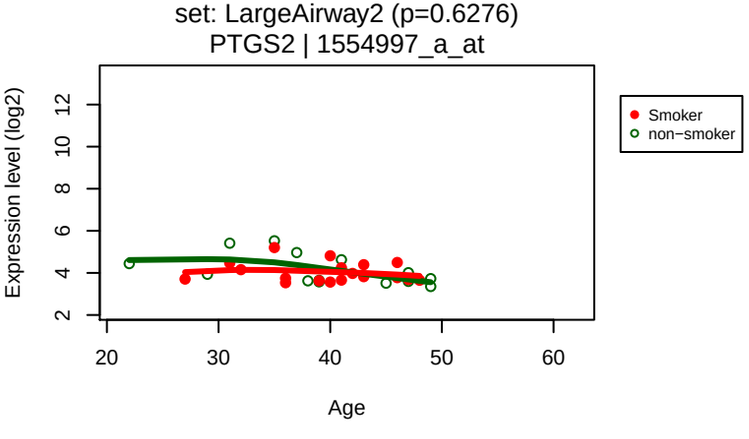

Gene expression ratio (smokers/nonsmokers)
gene: PTGS2 | 1554997_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7361
p=0.6257
p=0.7715
p=0.4333
p=0.6276
p=0.2055
p=0.757
p=0.1651
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

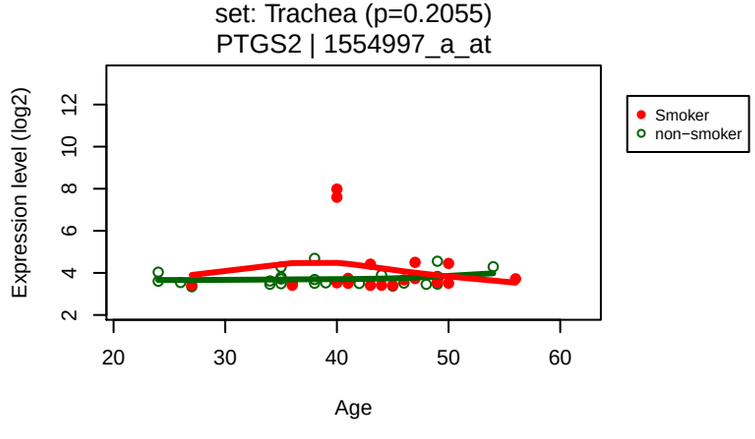

Gene expression ratio in Males
gene: PTGS2 | 1554997_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6007
p=0.4571
p=0.5043
p=0.1849
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGS2 | 1554997_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8099
p=0.4668
p=0.3642
p=0.7392
SmallAirway2
LargeAirway2
Trachea
Macrophage2

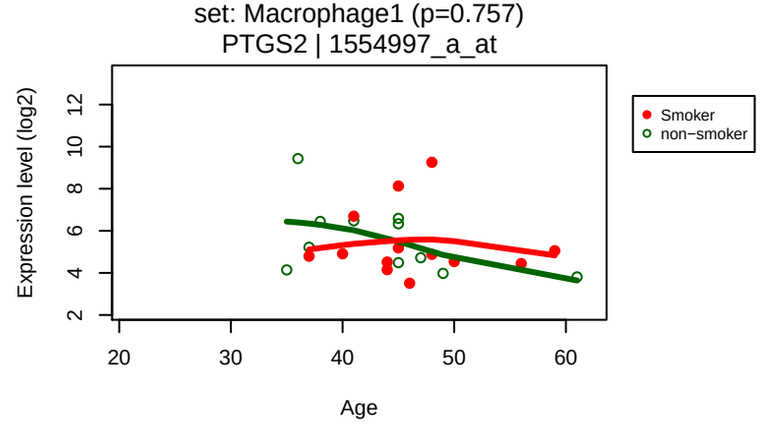

set: SmallAirway1 (p=0.7361)
PTGS2 | 1554997_a_at
Expression level (log2)
Age
Smoker
non−smoker

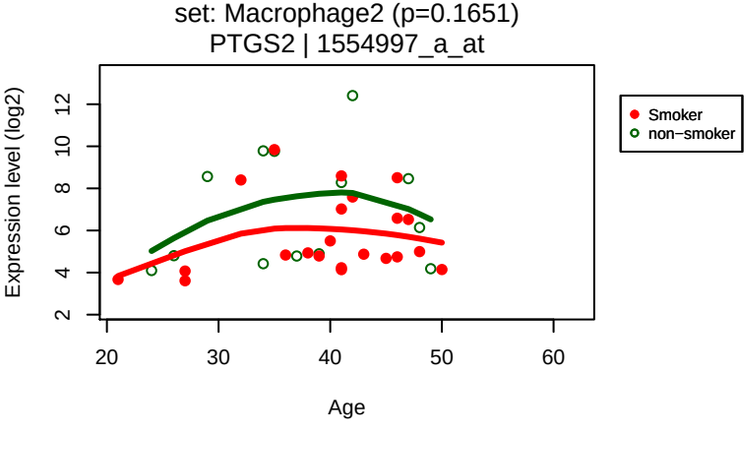

set: SmallAirway2 (p=0.6257)
PTGS2 | 1554997_a_at
Expression level (log2)
Age
Smoker
non−smoker

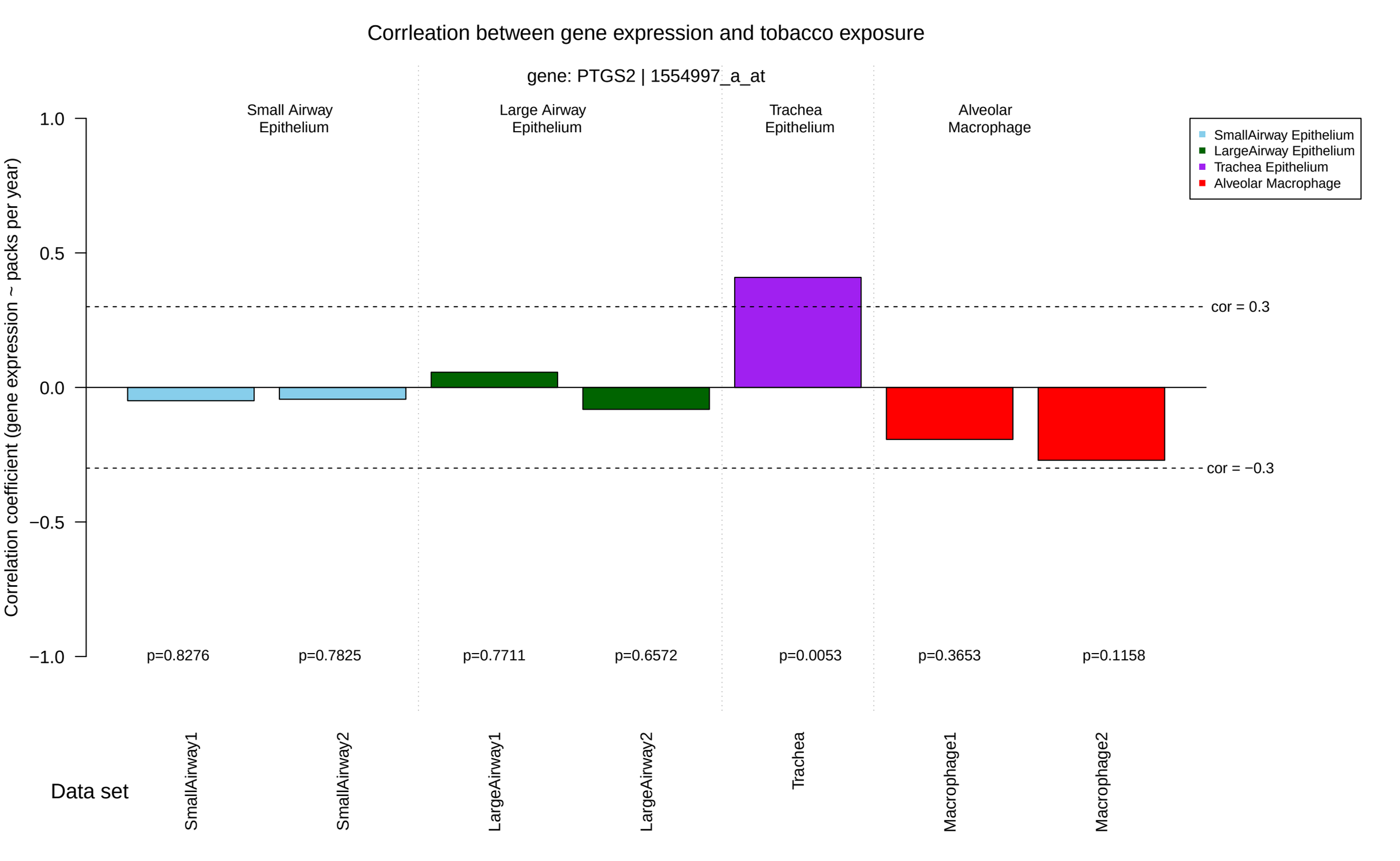

set: LargeAirway1 (p=0.4333)
PTGS2 | 1554997_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6276)
PTGS2 | 1554997_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2055)
PTGS2 | 1554997_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.757)
PTGS2 | 1554997_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1651)
PTGS2 | 1554997_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGS2 | 1554997_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8276
p=0.7825
p=0.7711
p=0.6572
p=0.0053
p=0.3653
p=0.1158
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2